

PROBLEMS

OBJECTIVE 10-1 EQUIVALENT IMPEDANCE (SECTS. 10-1, 10-2)

Given a linear circuit, use series and parallel equivalence to find the equivalent impedance at specified terminal pairs. Select element values to obtain specified pole locations. See Examples 10-2, 10-4, 10-5, 10-6, and Exercises 10-4, 10-5, 10-6, 10-7.

10-1 **D** For a series RC circuit find $Z_{EQ}(s)$ and then select R and C so that there is a pole at zero and a zero at -1 krad/s.

10-2 **D** For a parallel RC circuit find $Z_{EQ}(s)$ and then select R and C so that there is a pole at -10 krad/s.

10-3 **D** For the circuit of Figure P10-3:

(a) Find and express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

(b) Select values of R and C to locate a pole at -640 rad/s. Where is the resulting zero?

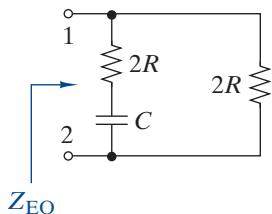


FIGURE P10-3

10-4 **D** For the circuit of Figure P10-4:

(a) Find and express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

(b) Select values of R and C to locate a zero at -33 krad/s.

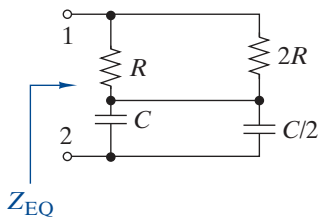


FIGURE P10-4

10-5 **D** For the circuit of Figure P10-5:

(a) Find and express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

(b) If $R = 2$ k Ω and $C = 0.1$ μ F, select a value of L to locate zeros at $\pm j5000$ rad/s.

(c) Where are the poles located once you have selected the inductor in part (b)?

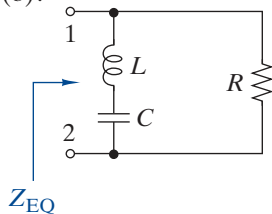


FIGURE P10-5

10-6 **D** For the circuit of Figure P10-6:

(a) Find and express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

(b) Select values of R and L to locate a pole at -150 rad/s. Where is the resulting zero?

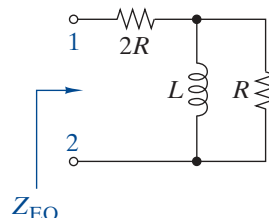


FIGURE P10-6

10-7 **D** For the circuit of Figure P10-7:

(a) Find and express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

(b) Select values of R and L to locate a pole at -2 krad/s. Where are the resulting zeros?

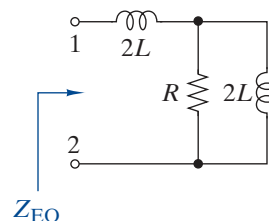


FIGURE P10-7

10-8 **D** For the circuit of Figure P10-8:

(a) Find and express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

(b) If $R = 10$ k Ω , select values of L and C to locate poles at $\pm j200$ krad/s. Where are the resulting zeros?

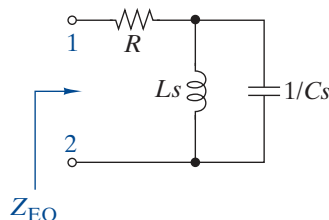


FIGURE P10-8

10-9 For the circuit of Figure P10-9:

(a) If $R = 1$ k Ω , $L = 2$ H, and $C = 0.5$ μ F locate the poles and zeros of $Z_{EQ}(s)$?

(b) If we were to increase the inductance to 5 H, how would the poles and zeros change?

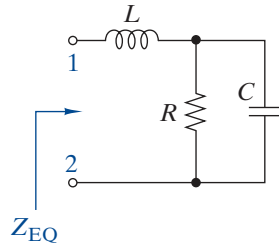


FIGURE P10-9

10-10 Find $Z_{EQ1}(s)$ and $Z_{EQ2}(s)$ for the bridge-T circuit in Figure P10-10. Express each impedance as a rational function and locate its poles and zeros.

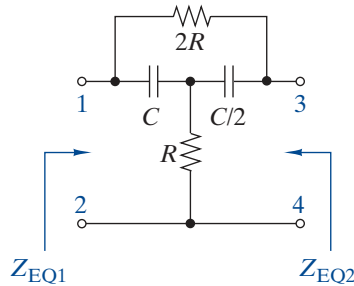


FIGURE P10-10

10-11 For the circuit of Figure P10-11:

- Find and express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.
- If its poles were located at -20 krad/s and zero, where would the poles move to if the value of R was reduced to 25% of its current value?

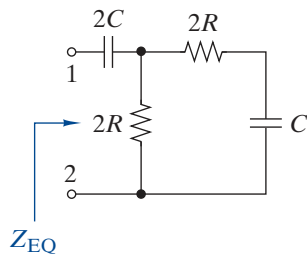


FIGURE P10-11

10-12  For the two-port circuit of Figure P10-12:

- Find $Z_{EQ1}(s)$ and Z_{EQ2} , and express each impedance as a rational function and locate its poles and zeros.
- Select values of R and L to place a pole at -500 Hz.

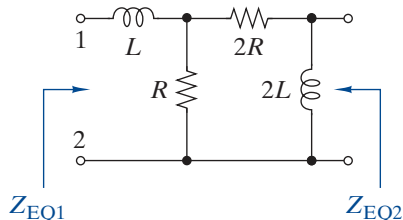


FIGURE P10-12

10-13  Find the equivalent impedance between terminals 1 and 2 in Figure P10-13. Select values of R and L so that

$Z_{EQ}(s)$ has a pole at $s = -3000$ rad/s. Locate the zeros of $Z_{EQ}(s)$ for your choice of R and L .

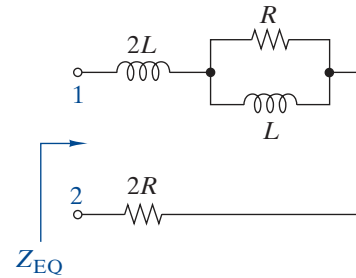


FIGURE P10-13

OBJECTIVE 10-2 BASIC CIRCUIT ANALYSIS TECHNIQUES (SECTS. 10-2, 10-3)

Given a linear circuit,

- Determine the initial conditions (if not given) and transform the circuit into the s domain.
- Solve for zero-state and zero-input responses using circuit reduction, the unit output method, Thévenin or Norton equivalent circuits, or superposition.
- Identify the forced and natural poles in the responses, or select circuit parameters to place the natural poles at specified locations.

See Examples 10-1, 10-2, 10-3, 10-4, 10-5, 10-7, 10-8, 10-9, 10-10, 10-11, 10-12, and Exercises 10-1, 10-2, 10-3, 10-4, 10-6, 10-7, 10-8, 10-9, 10-10, 10-11, 10-12, 10-13, 10-14.

10-14 For the circuit of Figure P10-14:

- Use voltage division to find $V_O(s)$.
- Use the look-back method to find $Z_T(s)$.

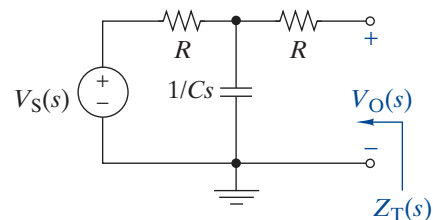


FIGURE P10-14

10-15 For the circuit of Figure P10-15:

- Use current division to find $I_2(s)$.
- Use the look-back method to find $Z_N(s)$.

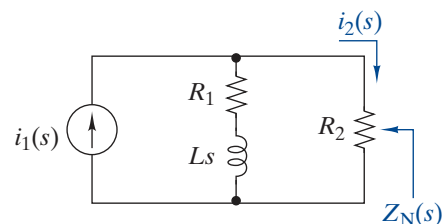


FIGURE P10-15

10-16  The circuit of Figure P10-16 needs to produce an output transform of

$$V_O(s) = \frac{V_A(RCs + 1)}{s(2RCs + 1)}$$

when the input is V_A/s . Design an appropriate circuit.

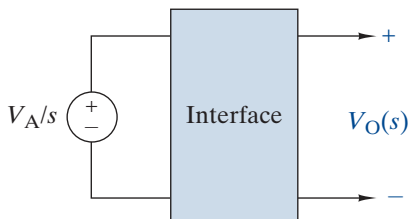


FIGURE P10-16

10-17  Find the Thévenin equivalent for the circuit in Figure P10-17. Then select values for R and L so that the Thévenin voltage has a pole at -12 krad/s.

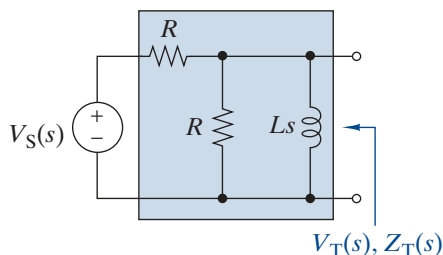


FIGURE P10-17

10-18 If the input to the RLC circuit of Figure P10-18 is $v_S(t) = u(t)$:

- Find the output voltage transform across each element.
- Compare the three outputs with regards to their respective poles and zeros.
- Use the initial and final value theorems to determine the value of the voltage across each element at $t = 0$ and $t = \infty$. What conclusions can one draw regarding the results?

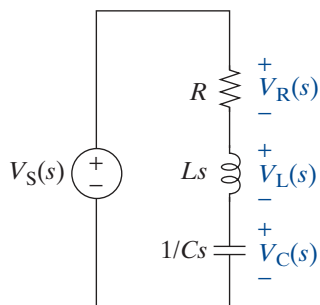


FIGURE P10-18

10-19 The switch in Figure P10-19 has been in position A for a long time and is moved to position B at $t = 0$. Transform the circuit into the s domain and solve for $I_L(s)$, $i_L(t)$, $V_O(s)$, and $v_O(t)$ in symbolic form.

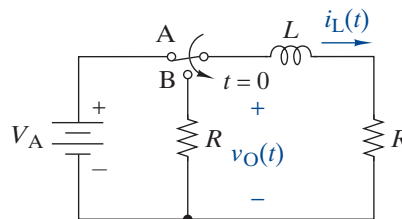


FIGURE P10-19

10-20 The switch in Figure P10-19 has been in position B for a long time and is moved to position A at $t = 0$. Transform the circuit into the s domain and solve for $I_L(s)$ and $i_L(t)$ in symbolic form.

10-21 The switch in Figure P10-21 has been in position A for a long time and is moved to position B at $t = 0$. Transform the circuit into the s domain and solve for $I_C(s)$, $i_C(t)$, $V_O(s)$, and $v_O(t)$ in symbolic form.

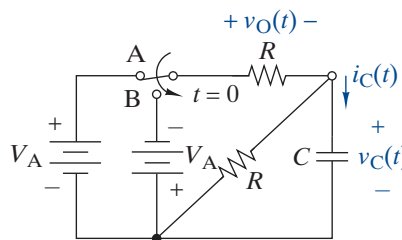


FIGURE P10-21

10-22 The switch in Figure P10-21 has been in position B for a long time and is moved to position A at $t = 0$. Transform the circuit into the s domain and solve for $V_C(s)$, $v_C(t)$, $V_L(s)$, and $v_O(t)$ in symbolic form.

10-23 Transform the circuit in Figure P10-23 into the s domain and find: $I_L(s)$, $i_L(t)$, $V_L(s)$, and $v_L(t)$ when $v_1(t) = V_A u(t)$ and $i_L(0) = I_A$.

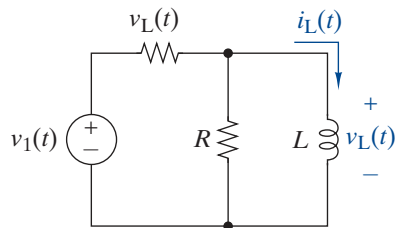


FIGURE P10-23

10-24 Transform the circuit in Figure P10-23 into the s domain and find $I_L(s)$ and $i_L(t)$ when $v_1(t) = V_A e^{-1000t} u(t)$, $R = 200 \Omega$, $L = 100$ mH, and $i_L(0) = 0$ A.

 **10-25** The switch in Figure P10-25 has been in position A for a long time and is moved to position B at $t = 0$.

- Transform the circuit into the s domain and solve for $I_L(s)$ in symbolic form.
- Repeat part (a) using MATLAB.
- Find $i_L(t)$ for $R_1 = R_2 = 500 \Omega$, $R_3 = 1$ k Ω , $L = 500$ mH, $C = 0.2$ μ F, and $V_A = 15$ V.

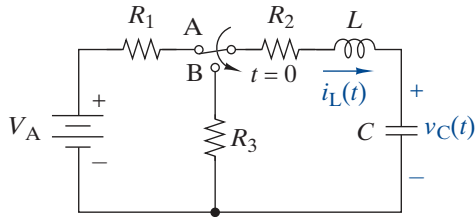


FIGURE P10-25

10-26 The switch in Figure P10-25 has been in position B for a long time and is moved to position A at $t = 0$.

(a) Transform the circuit into the s domain and solve for $V_C(s)$ in symbolic form.

(b) Find $v_C(t)$ for $R_1 = 50\ \Omega$, $R_2 = 250\ \Omega$, $R_3 = 1\ \text{k}\Omega$, $L = 500\ \text{mH}$, $C = 1\ \mu\text{F}$, and $V_A = 15\ \text{V}$. In addition, the inductor is not ideal, but has a parasitic resistance (in series) of $100\ \Omega$.

(c) Repeat part (b) using OrCAD.

10-27 The circuit in Figure P10-27 is in the zero state. The s -domain relationship between the input $I_1(s)$ and the output $I_R(s)$ is usually given as a ratio called a *network function*. Find $I_R(s)/I_1(s)$. Identify the poles and the zeros.

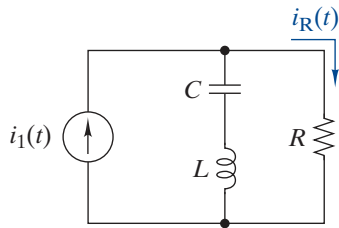


FIGURE P10-27

10-28 The circuit in Figure P10-28 is in the zero state. Find the s -domain relationship between the input $I_1(s)$ and the output $V_O(s)$. Identify the poles and the zeros.

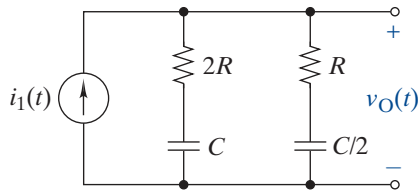


FIGURE P10-28

10-29 The initial conditions for the circuit in Figure P10-29 are $v_C(0) = 0$ and $i_L(0) = I_0$. Transform the circuit into the s domain and use superposition and voltage division to find the zero-state and zero-input components of $V_C(s)$.

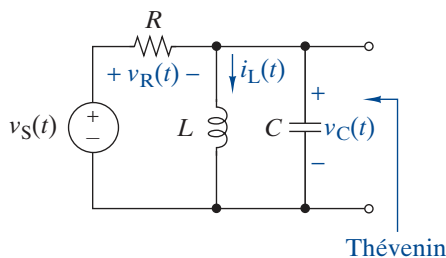


FIGURE P10-29

10-30 The initial conditions for the circuit in Figure P10-29 are $v_C(0) = 0$ and $i_L(0) = I_0$. Transform the circuit into the s domain and use superposition and voltage division to find the zero-state and zero-input components of $V_R(s)$.

10-31 The circuit in Figure P10-29 is in the zero state. Transform the circuit into the s domain and find the Thévenin equivalent circuit at the capacitor's terminals.

10-32 There is no energy stored in the capacitor in Figure P10-32 at $t = 0$. Transform the circuit into the s domain and use current division to find $v_O(t)$ when the input is $i_S(t) = 5e^{-500t}u(t)$ mA. Identify the forced and natural poles in $V_O(s)$.

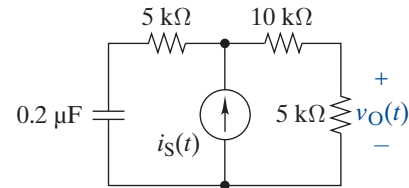


FIGURE P10-32

10-33 Repeat Problem 10-32 when $i_S(t) = 2.5 \cos 2000t u(t)$ mA.

10-34 The circuit in Figure P10-34 is in the zero state. Use a Thévenin equivalent to find the s -domain relationship between the input $I_S(s)$ and the interface current $I(s)$.

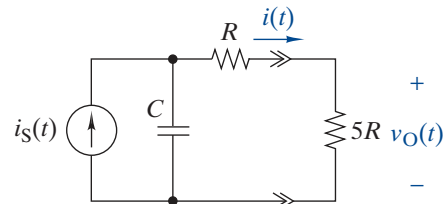


FIGURE P10-34

10-35 For the circuit of Figure 10-34:

(a) Find the Thévenin equivalent circuit that the $5R$ load resistor sees in when $v_C(0) = V_O$ V.

(b) Then find the voltage delivered to the load $v_O(t)$ if $v_C(0) = 10\ \text{V}$, $i_S(t) = 10u(t)$ mA, $R = 1\ \text{k}\Omega$, and $C = 2\ \mu\text{F}$.

(c) Identify the forced, natural, zero state, and zero input components of $v_O(t)$.

10-36 The circuit in Figure P10-36 is in the zero state. Find the Thévenin equivalent to the left of the interface.

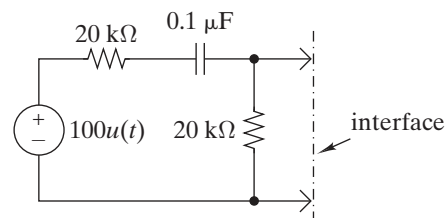


FIGURE P10-36

10-37  The Thévenin equivalent shown in Figure P10-37 needs to deliver

$$V_O(s) = \frac{50000}{(s+1500)(s+10)} \text{ V-s}$$

to a 1-k Ω load. Design an interface to allow that to occur.

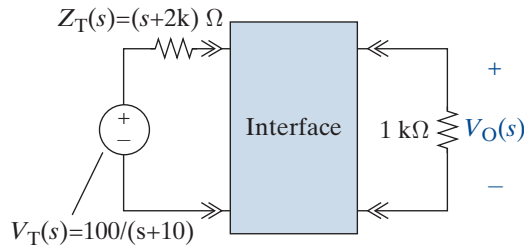


FIGURE P10-37

10-38 There is no initial energy stored in the circuit in Figure P10-38. Transform the circuit into the s domain and use superposition to find $V(s)$. Identify the forced and natural poles in $V(s)$.

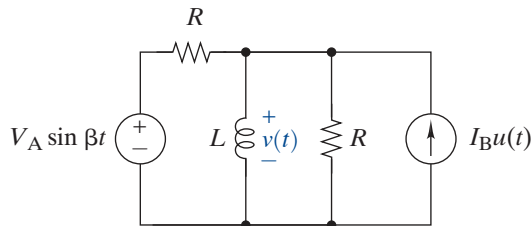


FIGURE P10-38

10-39 The equivalent impedance between a pair of terminals is

$$Z_{EQ}(s) = 2000 \left[\frac{s+3000}{s+2000} \right] \Omega$$

A voltage $v(t) = 10u(t)$ is applied across the terminals. Find the resulting current response $i(t)$.

10-40  There is no initial energy stored in the circuit in Figure P10-40. Use circuit reduction to find the output network function $V_2(s)/V_1(s)$. Then select values of R and C so that the poles of the network function are approximately -2618 and -382 rad/s.

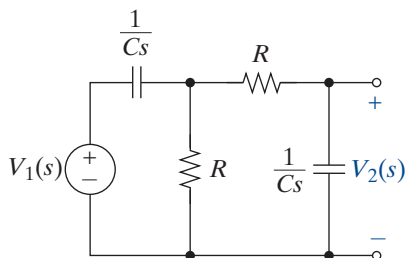


FIGURE P10-40

OBJECTIVE 10-3 GENERAL CIRCUIT ANALYSIS (SECTS. 10-4, 10-5, 10-6)

Given a linear circuit:

- Determine the initial conditions (if not given) and transform the circuit into the s domain.
- Solve for zero-state and zero-input response transforms and waveforms using node-voltage or mesh-current methods.
- Identify the forced and natural poles in the responses or select circuit parameters to place the natural poles at specified locations.

See Examples 10-13, 10-14, 10-15, 10-16, 10-17, 10-18, 10-19 and Exercises 10-15, 10-16, 10-17, 10-18, 10-19, 10-20, 10-21, 10-22, 10-23, 10-24.

10-41 There is no initial energy stored in the circuit in Figure P10-41.

- Transform the circuit into the s domain and formulate mesh-current equations.
- Show that the solution of these equations for $I_2(s)$ in symbolic form is

$$I_2(s) = I_B(s) = \frac{R_1 C s V_1(s)}{(R_1 + R_2) L C s^2 + (R_1 R_2 C + L) s + R_1}$$

- Identify the poles and zeros of $I_2(s)$.

- Find $i_2(t)$ for $v_1(t) = 10u(t)$ V, $R_1 = 1$ k Ω , $R_2 = 2$ k Ω , $L = 2$ H, and $C = 0.5$ μ F.

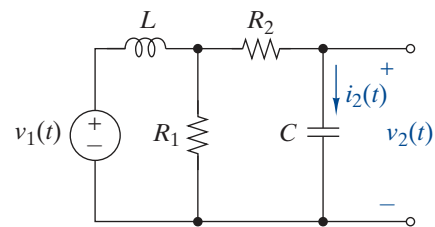


FIGURE P10-41

10-42 There is no initial energy stored in the circuit in Figure P10-41.

- Transform the circuit into the s domain and formulate node-voltage equations.
- Show that the solution of these equations for $V_2(s)$ in symbolic form is

$$V_2(s) = \frac{R_1 V_1(s)}{(R_1 + R_2) L C s^2 + (R_1 R_2 C + L) s + R_1}$$

- Identify the natural and forced poles of $V_2(s)$.

- Find $v_2(t)$ for $v_1(t) = 25u(t)$ V, $R_1 = R_2 = 500$ Ω , $L = 0.5$ H, and $C = 2$ μ F.

10-43  There is no initial energy stored in the circuit in Figure P10-43.

- Transform the circuit into the s domain and formulate node-voltage equations.

- (b) Solve these equations for $V_2(s)$ in symbolic form.
 (c) Insert an OP AMP buffer at point A and solve for $V_2(s)$ in symbolic form. How did inserting the buffer change the denominator, and therefore, the location of the poles?
 (d) Select values of R_1 , R_2 , C_1 , and C_2 to locate a pole at 10 krad/s and a second pole at 100 krad/s. You can choose either the circuit with or the one without the OP AMP for your design.

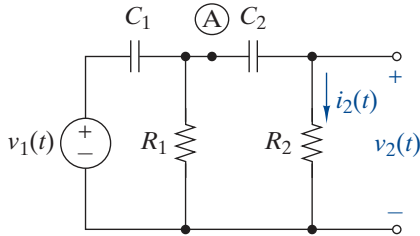


FIGURE P10-43

- 10-44 **D** There is no initial energy stored in the circuit in Figure P10-43. The Thévenin equivalent circuit to the left of point A when a unit step is applied is

$$V_T(s) = \frac{1}{s + 10^3} \text{ V-s, and } Z_T(s) = \frac{10^6}{s + 10^3} \Omega$$

Select values for R_2 and C_2 such that the output transform is

$$V_2(s) = \frac{s}{s^2 + 3000s + 10^6} \text{ V-s}$$

- 10-45 There is no initial energy stored in the bridged-T circuit in Figure P10-45.
 (a) Transform the circuit into the s domain and formulate mesh-current equations.
 (b) Use the mesh-current equations to find the s -domain relationship between the input $V_1(s)$ and the output $V_2(s)$.

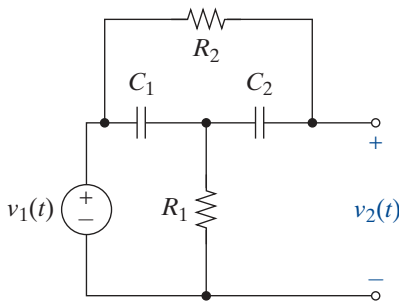


FIGURE P10-45

- 10-46 There is no initial energy stored in the bridged-T circuit in Figure P10-45.
 (a) Transform the circuit into the s domain and formulate node-voltage equations.
 (b) Use the node-voltage equations to find the s -domain relationship between the input $V_1(s)$ and the output $V_2(s)$.
 10-47 Find the transform of the Thévenin equivalent circuit looking into the $V_2(t)$ terminals for the circuit of P10-45.

- 10-48 There is no initial energy stored in the circuit in Figure P10-48.

- (a) Find the zero-state mesh currents $i_A(t)$ and $i_B(t)$ when $v_1(t) = 50 e^{-1000t} u(t)$ V.
 (b) Validate your answers using OrCAD.

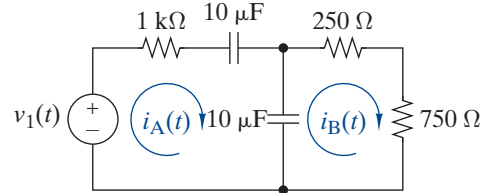


FIGURE P10-48

- 10-49 There is no external input in the circuit in Figure P10-49.

- (a) Find the zero-input node voltages $v_A(t)$ and $v_B(t)$, and the voltage across the capacitor $v_C(t)$ when $v_C(0) = -5$ V and $i_L(0) = 0$ A.
 (b) Use MATLAB to plot your results in (a).
 (c) Use OrCAD to validate your results in (a).
 (d) Compare the MATLAB and OrCAD plots. Are they the same?

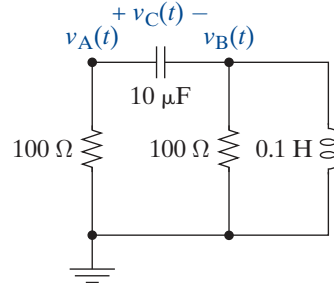


FIGURE P10-49

- 10-50 **D** The two-OP AMP circuit in Figure P10-50 is a band-pass filter.

- (a) Your task is to design such a filter so that the low-frequency cutoff is 1000 rad/s and the high-frequency cutoff is 100,000 rad/s. (Hint: See Example 10-16 and Exercise 10-18.)
 (b) Show that your design is correct using OrCAD.

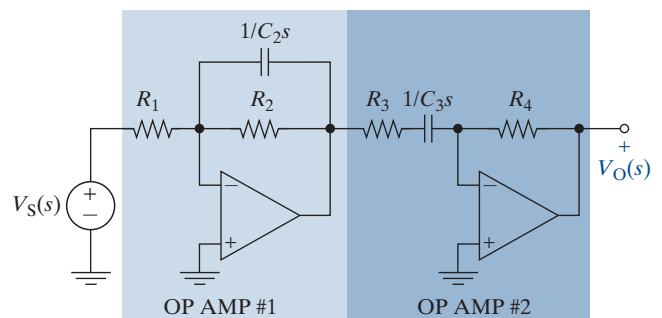


FIGURE P10-50

- 10-51 **D** The circuit in Figure P10-51 is in the zero state. Use node-voltage equations to find the circuit determinant.

Select values of R , C , and μ so that the circuit has $\omega_0 = 5$ krad/s and $\zeta = 0.707$.

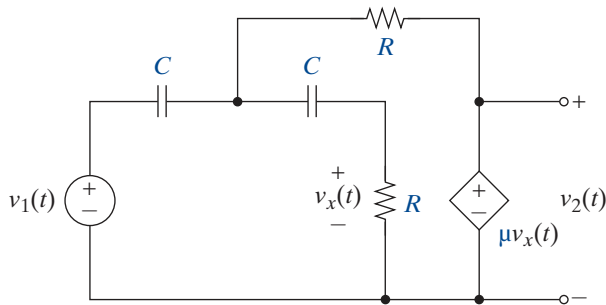


FIGURE P10-51

- 10-52 **D** The circuit in Figure P10-52 is in the zero state. Use mesh-current equations to find the circuit determinant. Select values of R , L , and C so that the circuit has $\omega_0 = 10$ krad/s and $\zeta = 1$.

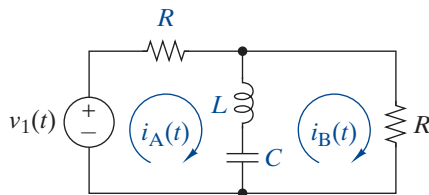


FIGURE P10-52

- 10-53 **D** The OP AMP circuit in Figure P10-53 is in the zero state. Use node-voltage equations to find the circuit determinant. Select values of R , C_1 , and C_2 so that the circuit has $\omega_0 = 10$ krad/s and $\zeta = 1$.

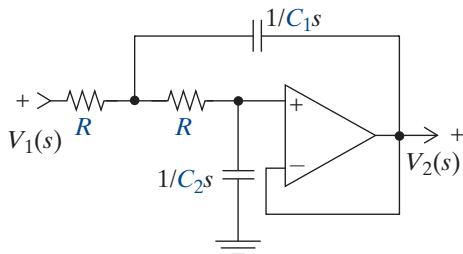


FIGURE P10-53

- 10-54 **E** Compare the results of your designs of the circuits in Figures P10-52 and P10-53. Since both circuits purport to have the same response characteristics, what are the advantages and disadvantages of each?

- 10-55 Three node voltages are shown in Figure P10-55.

- Explain why only one of the node voltages is independent.
- Write a node voltage equation in the independent node voltage.
- If $V_C(s)$ is the circuit's output, find the output-input ratio or *network function*, $V_C(s)/V_S(s)$.

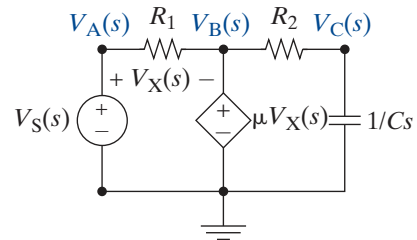


FIGURE P10-55

- 10-56 Three mesh currents are shown in Figure P10-56.

- Explain why only two of these mesh currents are independent.
- Write s -domain mesh-current equations in the two independent mesh currents.

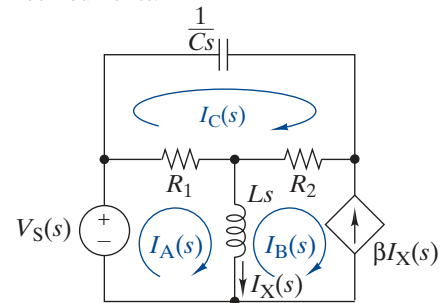


FIGURE P10-56

- 10-57 The switch in Figure P10-57 has been in position A for a long time and is moved to position B at $t = 0$.

- Write an appropriate set of node-voltage or mesh-current equations in the s -domain.
- Use MATLAB to solve for $V_C(s)$ and $v_C(t)$. Also using MATLAB, plot $v_C(t)$ and the exponential source on the same axes.
- Validate your results for $v_C(t)$ using OrCAD. Include the exponential source (See web Appendix C) in your Probe plot.

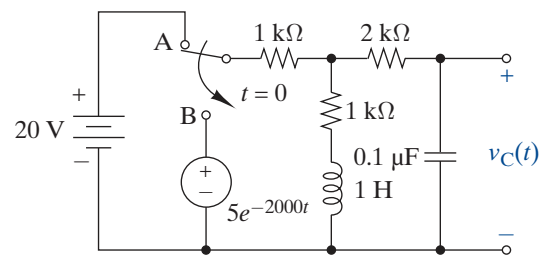


FIGURE P10-57

- 10-58 There is no energy stored in the circuit in Figure P10-58 at $t = 0$. Transform the circuit into the s domain and solve for $V_O(s)$ and $v_O(t)$.

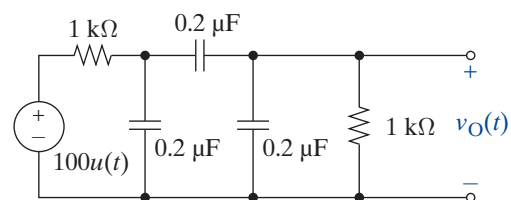


FIGURE P10-58

10–59 The switch in Figure P10–59 has been open for a long time and is closed at $t = 0$. Transform the circuit into the s domain and solve for $I_O(s)$ and $i_O(t)$

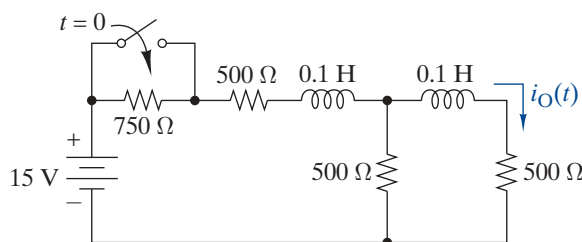


FIGURE P10–59

10–60 There is no initial energy stored in the circuit in Figure P10–60.

- (a) Transform the circuit into the s domain and solve for $V_O(s)$ in symbolic form.
 (b) If $R_1 = R_2 = 500\ \Omega$, select values of L and C to produce $\zeta = 0.707$ and $\omega_0 = 707\text{ rad/s}$. (Hint: Try values of L first, i.e., 100 mH, 200 mH, etc.)

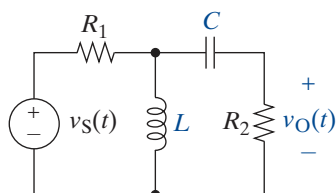


FIGURE P10–60

10–61 With the circuit in the zero state, the input to the integrator shown in Figure P10–61 is $v_1(t) = \cos 1000t\text{ V}$. The desired output is $v_2(t) = -\sin 1000t\text{ V}$. Use Laplace to select values of R and C to produce the desired output. If the capacitor had 10 V across it at $t = 0$, how would that affect the output?

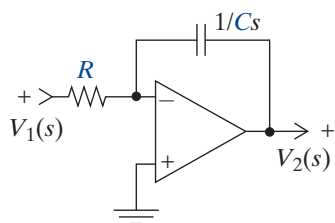


FIGURE P10–61

10–62 Show that the circuit in Figure P10–62 has natural poles at $s = -4/RC$ and $s = -2/RC \pm j2/RC$ when $L = R^2C/4$.

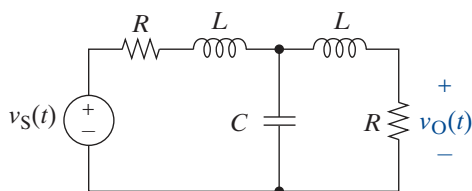


FIGURE P10–62

10–63 Find the range of the gain μ for which the circuit's output $V_X(s)$ in Figure P10–63 is stable (i.e., all poles are in the left hand side of the s plane.)

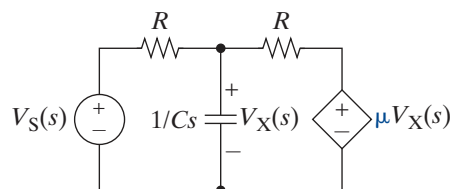


FIGURE P10–63

10–64 The circuit in Figure P10–64 is shown in the t domain with initial values for the energy storage devices.

- (a) Transform the circuit into the s domain and write a set of node-voltage equations.
 (b) Transform the circuit into the s domain and write a set of mesh-current equations.
 (c) With the circuit in the zero state, use symbolic operations in MATLAB to solve for the node voltages.

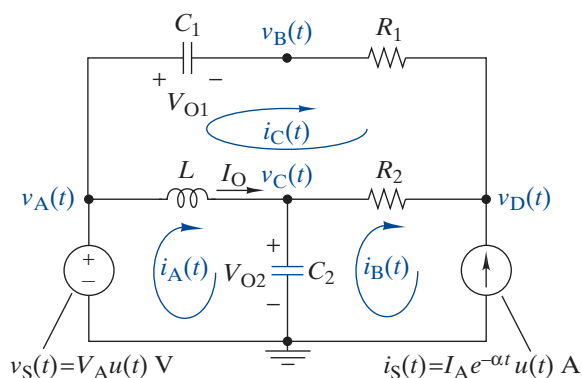


FIGURE P10–64

INTEGRATING PROBLEMS

10–65 Thévenin's Theorem from Time-domain Data

A black box containing a linear circuit has an on-off switch and a pair of external terminals. When the switch is turned on, the open-circuit voltage between the external terminals is observed to be

$$v_{OC}(t) = (10e^{-10t} - 10e^{-50t})u(t)\text{ V}$$

The short circuit current was observed to be

$$i_{SC}(t) = [(100e^{-10t} + 200e^{-50t} + 50e^{-100t})]u(t)\text{ mA}$$

A 50-Ω load resistance is connected across the terminals and the switch turned on again. What is the voltage delivered to the load?

10–66 Design a Load Impedance

In order to match the Thévenin impedance of a source, the load impedance in Figure P10–66 must be $Z_L(s) = \frac{s+5}{s+10}$.

- (a) What impedance $Z_2(s)$ is required?

(b) How would you realize $Z_2(s)$ using only resistors, inductors, and/or capacitors? (Hint: Write $Z_L(s)$ as a sum of admittances, then solve for $Y_2(s)$.)

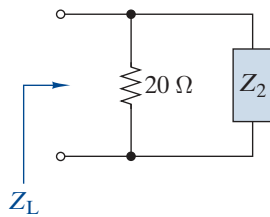


FIGURE P10-66

10-67 RC Circuit Analysis and Design A D

The RC circuits in Figure P10-67 represent the situation at the input to an oscilloscope. The parallel combination of R_1 and C_1 represent the probe used to connect the oscilloscope to a test point. The parallel combination of R_2 and C_2 represent the input impedance of the oscilloscope.

(a) Assuming zero initial conditions, transform the circuit into the s -domain and find the relationship between the test point voltage $V_S(s)$ and the voltage $V_O(s)$ at the oscilloscope's input.

(b) For $R_2 = 15 \text{ M}\Omega$ and $C_2 = 3 \text{ pF}$, determine the values of R_1 and C_1 that make the input voltage a scaled duplicate of the test point voltage.

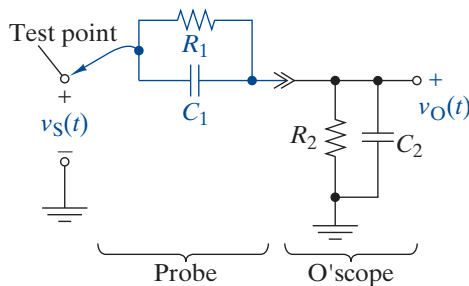


FIGURE P10-67

10-68 s-domain OP AMP Circuit Analysis A

The OP AMP circuit in Figure P10-68 is in the zero-state. Transform the circuit into the s -domain and use the OPAMP circuit analysis techniques developed in Section 4-4 to find the relationship between the input $V_1(s)$ and the output $V_2(s)$.

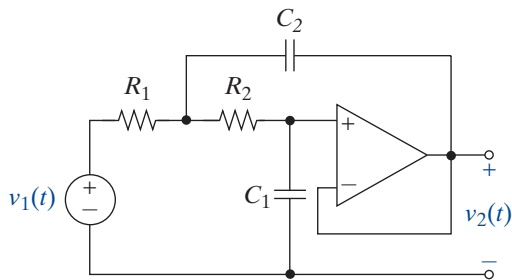


FIGURE P10-68

10-69 Pulse Conversion Circuit E

The purpose of the test setup in Figure P10-69 is to deliver damped sine pulses to the test load. The excitation comes from a 1-Hz square wave generator. The pulse conversion circuit must deliver damped sine waveforms with $\zeta < 0.5$ and $\omega_0 > 10 \text{ krad/s}$ to 50- Ω and 600- Ω loads. The recommended values for the pulse conversion circuit are $L = 10 \text{ mH}$ and $C = 0.1 \text{ }\mu\text{F}$. Verify that the test setup meets the specifications. (Hint: Compute the voltage across the load for an input signal equal to a unit step function, $u(t)$, and then again for a negative unit step function, $-u(t)$.) Note that the output of a square wave generator is the sum of a series of step functions.

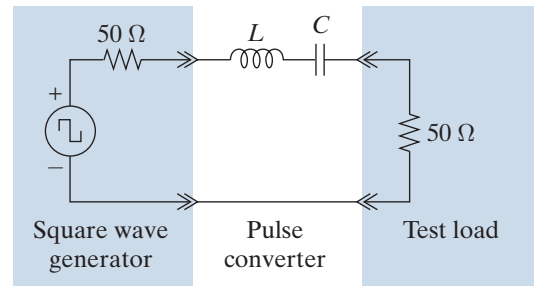
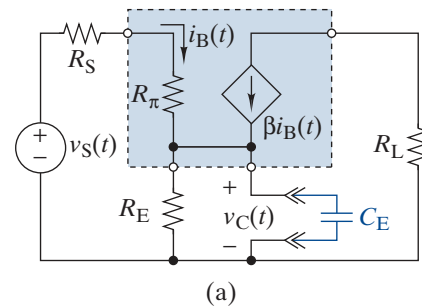


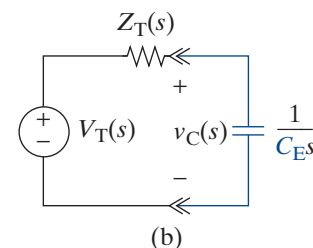
FIGURE P10-69

10-70 By-Pass Capacitor Design D

In transistor amplifier design, a by-pass capacitor is connected across the emitter resistor R_E to effectively short out the emitter resistor at signal frequencies. This then, improves the gain of the transistor for the desired ac signals. The circuit in Figure P10-70(a) is a common-emitter amplifier. The shaded portion is a low-frequency model of the transistor in use. In this problem the task is to design a proper by-pass capacitor so that there is a pole at $s = -300 \text{ rad/s}$. Reduce the circuit to its Thévenin equivalent as shown in Figure P10-70(b), and then select the proper capacitor. $R_S = 12 \text{ k}\Omega$, $R_\pi = 2.5 \text{ k}\Omega$, $R_E = 3.3 \text{ k}\Omega$, $R_L = 1 \text{ k}\Omega$, and $\beta = 50$.



(a)



(b)

FIGURE P10-70